V.L Memorial Public School

475 Vijay Nagar

FRACTIONS IN DISGUISE

Class 08 - Maths (Ganita Prakash)

Section A

1. Which is equal to 15% of 200?

   a) 30% of 100 b) 10% of 250

   c) 20% of 120 d) 5% of 400

2. If 40% of a number is 120, what is the number?

   a) 360 b) 240

   c) 200 d) 300

3. Convert the fraction 3/5 into a percentage.

   a) 30% b) 50%

   c) 75% d) 60%

4. A school has 600 students.
   12% of them are in Class VIII.
   How many students are in Class VIII?

   a) 96 b) 84

   c) 60 d) 72

5. What is 50% of 84?

   a) 48 b) 42

   c) 21 d) 52

6. 20% of 200 is:

   a) 40 b) 30

   c) 60 d) 20

7. Which fraction is equal to 45%?

   a) $\frac{5}{9}$ b) $\frac{9}{20}$

   c) $\frac{20}{9}$ d) $\frac{45}{10}$

8. The compound interest on ₹ 5000 at 4% per annum for 2 years compounded annually is

   a) ₹ 708 b) ₹ 408

   c) ₹ 804 d) ₹ 5408

9. Lata bought a teapot for ₹ 120 and a set of cups for ₹ 400. She sold the teapot at a profit of 5% and the cups at a loss of 5%. The amount received by her is

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a) ₹ 534 b) ₹ 546

c) ₹ 494 d) ₹ 506

10. A shopkeeper has 50 bulbs, out of which 45 are non-defective. The percent of defective bulbs is

a) 4% b) 10%

c) 8% d) 12%

11. ₹ 7000 is given at 7% p.a. rate of interest. The interest which will be received at the end of the three years is

a) ₹ 1410 b) ₹ 1400

c) ₹ 1470 d) ₹ 1455

12. In a mixture, the ratio of flour to sugar is 3 : 1.
What percentage of the mixture is sugar?

a) 33.33% b) 20%

c) 25% d) 30%

13. A number is increased by 20%.
Which expression represents the new number if the original number is x?

a) x + 0.2 b) 0.2x

c) x - 0.2x d) 1.2x

14. Sagun lends ₹ 2,88,000 to her friend at the rate of 12% p.a. for 9 months. How much interest will she get?

a) ₹ 25,920 b) ₹ 3,11,040

c) ₹ 34,560 d) ₹ 3,13,920

15. A shirt is sold for ₹990 including 10% GST.
What is the marked price of the shirt?

a) ₹910 b) ₹980

c) ₹900 d) ₹1000

16. The price of a bag is ₹800. A discount of 10% is given. What is the selling price?

a) ₹700 b) ₹720

c) ₹780 d) ₹740

17. If an article bought for ₹ 1357 includes 18% GST, then the price of the article without GST is

a) ₹ 1250 b) ₹ 1150

c) ₹ 1126 d) ₹ 1162

18. 0.4 written as a percentage is:

a) 400% b) 40%

c) 0.4% d) 4%

19. Which of the following fractions is equal to 25%?

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a) 1/4 b) 3/4

c) 2/5 d) 1/5

20. Which of the following is the correct method to convert a fraction to a percentage?

a) Divide the fraction by 100 b) Multiply the fraction by 100

c) Multiply the denominator by 100 only d) Add 100 to the fraction

21. **Assertion (A):** Simple interest on an amount of money for 2 years at 10% pa is less than compound interest on the same sum for the same time and rate.
Reason (R): If the principal is doubled, the interest also gets doubled.

a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false. d) A is false but R is true.

22. **Assertion (A):** Neha bought a cup for ₹76 after getting a 5% discount on its marked price of ₹80.
Reason (R): Sale price = Marked price - Discount.

a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false. d) A is false but R is true.

23. **Assertion (A):** ₹ 1600 amounts to ₹ 1681 in two years at a compound interest of 5% per annum, compounded annually.
Reason (R): Amount when interest is compounded annually is $A = P(1 + r)^t$.

a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false. d) A is false but R is true.

24. **Assertion (A):** 20% of 150 is equal to 30.
Reason (R): 20% of any quantity means multiplying the quantity by $\frac{20}{100}$.

a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false. d) A is false but R is true.

25. **Assertion (A):** An increase of 20% followed by a decrease of 20% always brings the value back to the original value.
Reason (R): Percentage change is calculated on the current value each time.

a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false. d) A is false but R is true.

26. **Assertion (A):** 50% is equal to the fraction $\frac{1}{2}$.
Reason (R): To convert a percentage into a fraction, we multiply it by 100.

a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.

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c) A is true but R is false.

d) A is false but R is true.

27. **Assertion (A):** $\frac{2}{5} = 40\%$

    Reason (R): To convert a fraction into a percentage, we multiply it by 100.

    a) Both A and R are true and R is the correct explanation of A.

    b) Both A and R are true but R is not the correct explanation of A.

    c) A is true but R is false.

    d) A is false but R is true.

28. **Assertion (A):** A discount of 30% + 20% is offered in a clearance sale. This is the same as a discount of 50%.

    Reason (R): Price after discount is always less than the marked price.

    a) Both A and R are true and R is the correct explanation of A.

    b) Both A and R are true but R is not the correct explanation of A.

    c) A is true but R is false.

    d) A is false but R is true.

29. **Assertion (A):** If 25% of a number is 80, then the number is 320.

    Reason (R): 25% is equal to $\frac{1}{4}$.

    a) Both A and R are true and R is the correct explanation of A.

    b) Both A and R are true but R is not the correct explanation of A.

    c) A is true but R is false.

    d) A is false but R is true.

30. The marked price of a book is ₹250. A discount of 12% is given. Find the selling price.
31. A number is increased by 10% and then decreased by 10%.
    Find the final value if the original number is 200.
32. If 15% of a number is 90, find the number.
33. Rainforests are home to 90,000 of the 2,50,000 identified plant species in the world. What percentage of the world's identified plant species are found in rainforests?
34. In a school, 45% of 600 students are girls. Find the number of girls.
35. The percentage of pure gold in 14-carat gold is about 58%. A 14-carat gold ring weighs 20 grams. How many grams of pure gold are in the ring?
36. Find 35% of 240.

Section B

37. A principal gives ₹ 3200 as simple interest at the rate of 10% p.a. in 2 years. What will be the compound interest at the rate of 5% p.a. for 3 years on the same principal?
38. The height of the tree has increased by 10% from last year. If its present height is 11m, what was its height last year?
39. If 30% of a number is 180, find the number and also find 45% of that number.
40. In what time will ₹ 16,000 amount to ₹ 20,800 at 10% per annum without compounding?
41. The population of a city increases at the rate of 5% pa. If the population in 2024 was 24,00,000, find the population in 2023 and 2025.
42. A machine was purchased 3 years ago at ₹ x and its value depreciated by 10% per annum. If its present value is ₹ 1,21,743, what is the value of x?
43. A student scored 54 marks out of 75. Find the percentage of marks and decide whether the student scored more than 70%.

44. The price of a commodity has increased by 12%. If it used to cost ₹ 200 earlier, what does it cost now?

Section C

45. A recipe requires $2\frac{3}{4}$ cups of flour. If you want to make half of the recipe, how much flour do you need?
46. In an examination, Rahul scored 432 marks out of 600. What percentage did he score? If the passing percentage is 40%, by what percentage did he exceed the passing marks?
47. Priya completed $\frac{3}{5}$ of her homework on Monday and $\frac{1}{4}$ on Tuesday. What fraction of her homework is still incomplete?
48. A farmer sold $\frac{2}{5}$ of his harvest to one buyer and $\frac{1}{3}$ to another. If he had 600 kg remaining, what was the total harvest?
49. A piece of ribbon is $6\frac{3}{4}$ meters long. It is cut into pieces of $\frac{3}{4}$ meter each. How many pieces can be obtained?
50. The population of a town increases by 8% in the first year and decreases by 5% in the second year. If the population at the end of 2 years is 82,080, what was the original population?
51. A water tank is $\frac{2}{3}$ full. If 120 liters of water is added, the tank becomes $\frac{5}{6}$ full. What is the total capacity of the tank?
52. CP of 10 items is the same as SP of 8 items. Find the profit or loss percent.
53. A man invested ₹15,000 in a bank at 10% per annum compound interest. After 2 years, he withdrew a certain amount and the remaining amount earned interest for one more year, becoming ₹13,310. How much did he withdraw after 2 years?
54. In a laboratory, the number of bacteria in a certain experiment increases at the rate of 2.5% per hour. Find the number of bacteria at the end of 2 hours if the initial count is 5,06,000.

Section D

55. **Read the following text carefully and answer the questions that follow:**
    Eesha and her friends are discussing their test results. Eesha scored 42 marks out of 50 in an English test and 70 marks out of 80 in a Science test. Eesha thinks she did better in English because she lost only 8 marks there compared to 10 marks in Science. Her friend Reema disagrees, saying that since Eesha scored more marks in Science, she performed better there. Another friend Vishu suggests they cannot compare because the maximum marks are different. Their teacher explains that converting scores to percentages makes comparison easier and fairer.

    Questions:

    i. What is Eesha's percentage score in English? **(1)**
    ii. What is Eesha's percentage score in Science? **(1)**
    iii. In which subject did Eesha perform better, and by how much percentage? **(2)**

    OR

    If the passing percentage is 80% in both subjects, did Eesha pass both? Justify your answer. **(2)**

56. **Read the following text carefully and answer the questions that follow:**
    Surya is an artist who wants to create a deep orange colour to paint a sunset. He mixes red and yellow paint to make this colour. In his mixture, red paint makes up 3/4 of the total volume. He needs to understand this as a percentage to explain his technique to his art students. Later, he decides to make 60 ml of this deep orange paint for a demonstration. His students are curious about how much red paint he actually uses. Surya explains that percentages are just fractions with denominator 100, and shows

them different methods to convert fractions to percentages using equivalent fractions and multiplication by 100.

Questions:

i. What percentage of Surya's paint mixture is red? **(1)**

ii. What percentage of the mixture is yellow? **(1)**

iii. If Surya makes 60 ml of deep orange paint and wants to create a lighter shade where red makes up only 60% of the mixture, how much red paint and yellow paint should he use? **(2)**

OR

In a different shade, the ratio of red to yellow paint is 3:2. What percentage of the mixture is red and what percentage is yellow? **(2)**

57. A recipe for the dish, halwa, for 4 people has the following ingredients in the given proportions- Rava: 40%, Sugar: 40%, and Ghee: 20%.

i. If you want to make halwa for 8 people, what is the proportion of each of the above ingredients?

ii. If the total weight of the ingredients is 2 kg, how much rava, sugar and ghee are present?

Section E

58. A fruit seller sells apples at a profit of 20%. If he had bought them at 10% less and sold for ₹5 more per kg, he would have gained 40%. Find the original cost price per kg.

59. A sum of ₹8000 is invested partly at 5% and partly at 8% simple interest per annum. If the total interest after 2 years is ₹1140, find how much was invested at each rate.

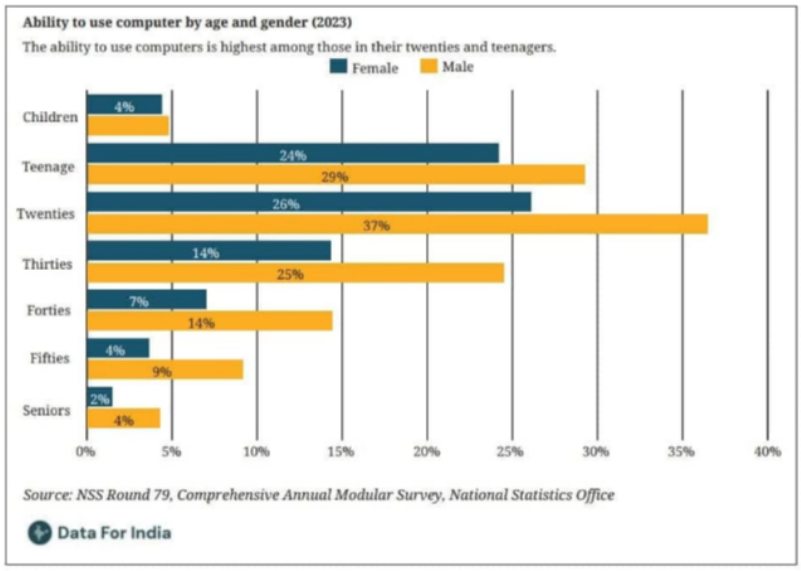
60. A mixture contains milk and water in the ratio 5:2. If 14 litres of water is added to the mixture, the percentage of water becomes 40%. Find the initial quantity of milk in the mixture.

61. A farmer had 800 kg of rice. He sold 35% of it at ₹40 per kg and the remaining at ₹45 per kg. Find:

i. The total amount he received

ii. If his total cost was ₹30,000, what was his profit percentage?

62. Look at the following graph.



Based on the graph, which of the following statement(s) are valid?

i. People in their twenties are the most computer-literate among all age groups.

ii. Women lag behind in the ability to use computers across age groups.

iii. There are more people in their twenties than teenagers.

iv. More than a quarter of people in their thirties can use computers.

v. Less than 1 in 10 aged 60 and above can use computers.

vi. Half of the people in their twenties can use computers.

63. A bank offers an interest rate of 8% per annum on fixed deposits. Ravi deposits ₹25,000 for 3 years. Calculate:

    i. The simple interest (without compounding)

    ii. The compound interest (with annual compounding)

    iii. How much more interest does he earn with compounding?

64. A container has a mixture of milk and water in the ratio 5:3. If 16 liters of the mixture is removed and replaced with water, the ratio becomes 3:5. Find:

    a. The initial quantity of milk in the container

    b. The initial quantity of water in the container

    c. The total initial quantity of the mixture

    d. The final quantity of milk after replacement

65. A bakery offers two discount schemes on a cake priced at ₹1,200:

    Scheme A: 30% + 20% successive discounts

    Scheme B: Flat 50% discount

    i. Calculate the final price under each scheme.

    ii. Which scheme is better for the customer?

    iii. What is the difference in the final prices?

    iv. Express the effective discount percentage for Scheme A.

66. The post office offers an interest of 7% p.a. How much interest would one get if one invests ₹ 50,000 for 3 years without compounding? How much more would one get if it was compounded?

67. Three friends - Amit, Bharat, and Chetan - contributed to buy a gift. Amit contributed $\frac{2}{5}$ of the total amount, Bharat contributed $\frac{1}{3}$ of the total amount, and Chetan contributed the remaining Rs. 260.

    i. What fraction of the total amount did Chetan contribute?

    ii. What was the total cost of the gift?

    iii. How much did Amit contribute?

    iv. How much did Bharat contribute?

Section F

68. **Fill in the blanks:**

    (a) 30% of k is 70, 60% of k is _____, 90% of k is _____, 120% of k is _____.

    (b) If 25% of a number is 100, the number is _____.

    (c) 5% GST is charged on an article marked ₹ 2000 after allowing a discount of 5%, then the amount payable is _____.

    (d) 100% of m is 215, 10% of m is _____, 1% of m is _____, 6% of m is _____.

    (e) 25% of 320 is _____.

    (f) _____ is 40% of 4.

    (g) 20% of 250 is _____.

    (h) If the price of an article is reduced by 20% and becomes ₹640, the original price was ₹_____.